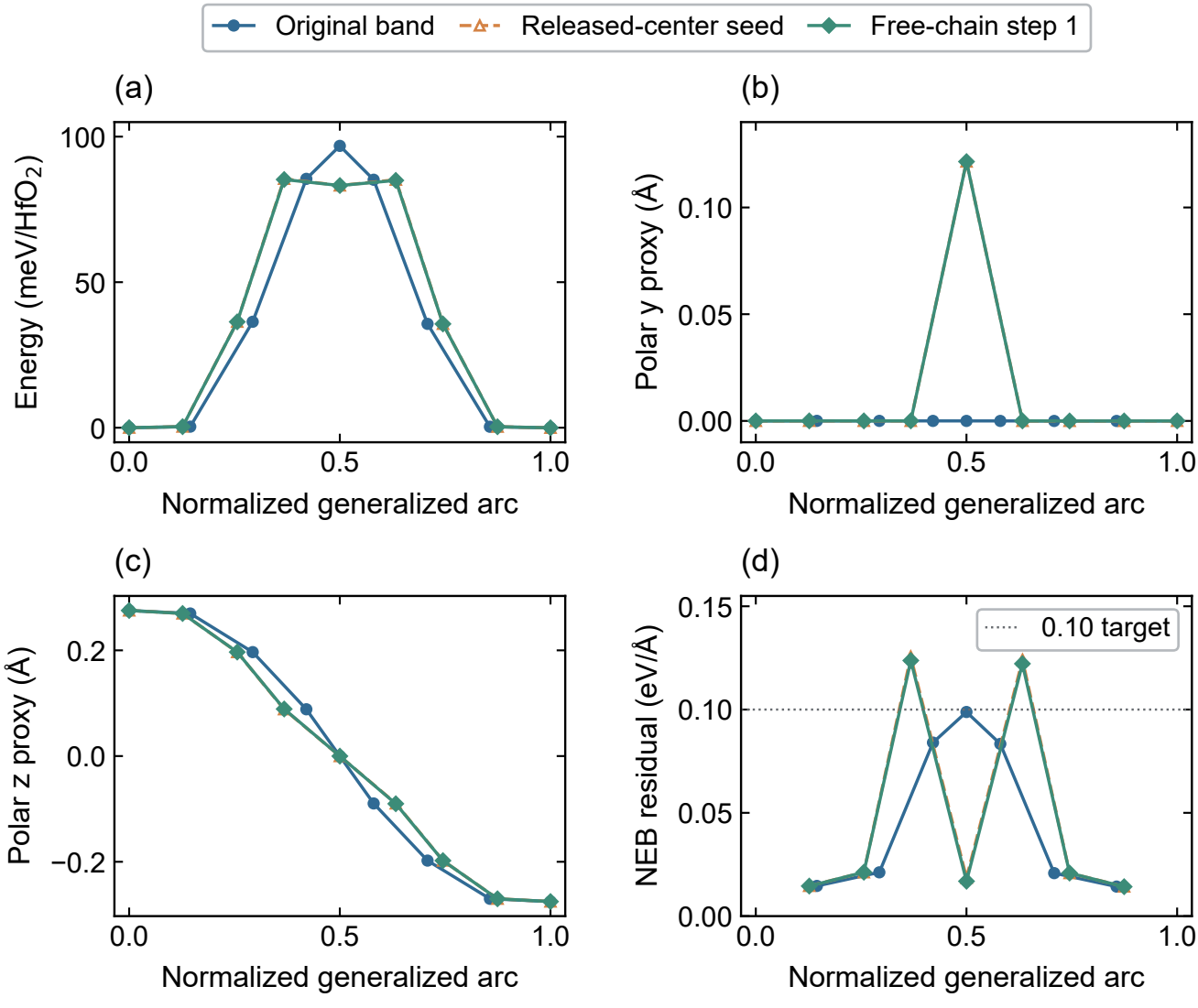
Original band
Released-center seed
Free-chain step 1
(a)
Energy (meV/HfO2)
100
50
0
0.0
0.5
1.0
Normalized generalized arc
(b)
Polar y proxy (Å)
0.10
0.05
0.00
0.0
0.5
1.0
Normalized generalized arc
(c)
Polar z proxy (Å)
0.2
0.0
−0.2
0.0
0.5
1.0
Normalized generalized arc
(d)
NEB residual (eV/Å)
0.15
0.10
0.05
0.00
0.10 target
0.0
0.5
1.0
Normalized generalized arc